Notes for the course in

FUNCTIONAL ANALYSIS

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0.1 Introduction

1. A C^* algebra satisfies the following conditions

   (1) $\|x+y\| \leq \|x\| + \|y\|$
   (2) $\|xy\| \leq \|x\|\|y\|$
   (3) $(x^*)^* = x$
   (4) $(x+y)^* = x^* + y^*$
   (5) $(\alpha x)^* = \overline{\alpha} x^*$
   (6) $(xy)^* = y^* x^*$
   (7) $\|x^*\| = \|x\|$
   (8) $\|x^* x\| = \|x\|^2$ (C^* property)

2. States are positive linear functionals. This corresponds to the expected value of the observable.

3. Cauchy-Schwarz Inequality:

$$\omega[(\alpha f + \beta g)^*(\alpha f + \beta g)] = \begin{pmatrix} \overline{\alpha} & \overline{\beta} \end{pmatrix} \begin{pmatrix} \omega(f^* f) & \omega(f^* g) \\ \omega(g^* f) & \omega(g^* g) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \geq 0$$

   and thus

$$|\omega(f^* g)| \leq \sqrt{\omega(f^* f)}\sqrt{\omega(g^* g)}.$$

4. We have, from the positivity $\omega[(\lambda A + \mathbb{1})^*(\lambda A + \mathbb{1})] \leq 0$, that $\omega(A^*) = \overline{\omega(A)}$.

   Proof: First, we expand to find that

$$0 \leq |\lambda|^2 \omega(A^* A) + \lambda \omega(A) + \overline{\lambda}\omega(A^*) + \omega(\mathbb{1}).$$

   First, we take $\lambda \in \mathbb{R}$ and note that this means that $\omega(A) + \omega(A^*)$ is real. Then, let λ be pure imaginary and see that $\omega(A) - \omega(A^*)$ is purely imaginary. Thus, we see that $\omega(A) = \overline{\omega(A^*)}$

0.2 C^* Algebras

1. Let $\mathcal{A}$ be a C^* algebra without identity. We can construct an algebra with identity $\overline{\mathcal{A}} = \mathbb{C} \times \mathcal{A}$ with identity using the product and sum

$$\begin{aligned} \mu(\alpha, A) + \lambda(\beta, B) &= (\mu\alpha + \lambda\beta, \mu A + \lambda B) \\ (\mu, A)(\beta, B) &= (\alpha\beta, \alpha B + \beta A + AB) \\ (\mu, A)^* &= (\overline{\alpha}, A^*) \end{aligned}$$

   and norm $\|(\alpha, A)\| = \sup_{\|B\|=1} \|\alpha B + AB\|$. The identity is $(1, 0)$.

2. Ideals are subalgebras.

3. Self adjoint ideals are two sided.

4. Factor spaces are algebras.

5. Two sided ideals are not necessarily self-adjoint! Let $\mathcal{A} = C(S^2)$ with product $(fg)(x) = f(x)g(x)$ and inversion $f^*(x) = \overline{f}(-x)$. Then $\mathcal{I} = \{f | f(\text{north pole}) = 0\}$. IT is clear that $\mathcal{I}$ is two sided and $\mathcal{I}^* = \{f | f(\text{south pole}) = 0\}$.

6. The resolvent of $A \in \mathcal{A}$ is the subset of $\mathbb{C}$ such that $\lambda\mathbb{1} - A$ is invertible for $\lambda \in r_{\mathcal{A}}(A)$.

7. The spectrum is the complement of the resolvent.

8. **(Von Neumann Series)** Formally, we have

$$\frac{1}{\lambda\mathbb{1} - A} = \frac{1}{\lambda} \sum_{m=0}^{\infty} \left(\frac{A}{\lambda}\right)^m.$$

   It is clear that this is a power series in $\frac{1}{\lambda}$ and converges for $\|A\|/|\lambda| < 1$. Here, we have used $\|A^n\| \leq \|A\|^n$. Hence, we cannot find the radius of convergence using the geometric series formula.

9. By a similar power series argument, we can show that the resolvent is open. Hence, the spectrum is compact.

10. The spectral radius is $R = \sup_{\lambda \in \sigma_{\mathcal{A}}(A)} |\lambda|$. We have $R \leq \|A\|$.

11. The spectral radius is given by $\rho(A) = \inf_{n \in \mathbb{N}} \|A^n\|^{1/n} = \lim_{n\to\infty} \|A^n\|^{1/n}$.

    This is the root test: The radius of convergence of $\sum_{n=0}^{\infty} A_n(z-a)^n$ is $\left(\limsup_{n\to\infty} \|A_n\|^{1/n}\right)^{-1}$.

12. In a unital *-algebra,

$$\sigma_{\mathcal{A}}(A) + \mu = \sigma_{\mathcal{A}}(A + \mu\mathbb{1}),\ \sigma_{\mathcal{A}}(A^*) = \overline{\sigma_{\mathcal{A}}(A)},\ \sigma_{\mathcal{A}}(AB) \cup \{0\} = \sigma_{\mathcal{A}}(BA) \cup \{0\}.$$

    The third statement is proven by $(\mathbb{1}\lambda + AB)A = A(\lambda\mathbb{1} + BA)$ and hence $(\lambda\mathbb{1} + AB)A\frac{1}{\lambda\mathbb{1}+BA} = A$, and then adding $(\lambda\mathbb{1} - AB)$ to get an inverse.

13. Let $\mathcal{A}$ be a unital C^* algebra.

    (a) If A is normal then $\rho(A) = \|A\|$.

$$\|A^{2^n}\|^2 = \|(A^*)^{2^n} A^{2^n}\| = \|(A^*A)^{2^n}\| = \|(A^*A)^{2^{n-1}}\|^2 = \|A^*A\|^{2^n} = \|A\|^{2^{n+1}}.$$

(b) If A is an isometry, i.e. $A^*A = \mathbb{1}$, then $\rho(A) = 1$.

$$\|A^n\|^2 = \|(A^*)^n A^n\| = \|\mathbb{1}\| = 1.$$

(c) If A is unitary, then $\sigma_{\mathcal{A}} \subseteq S^1$.

$$\sigma_{\mathcal{A}}(A) = \overline{\sigma_{\mathcal{A}}(A^*)} = \overline{\sigma_{\mathcal{A}}(A^{-1})} = \left(\overline{\sigma_{\mathcal{A}}(A)}\right)^{-1}.$$

(d) If A is self adjoint, then $\sigma_{\mathcal{A}}(A) \subseteq [-\|A\|, \|A\|] \subseteq \mathbb{R}$, and $\sigma_{\mathcal{A}}(A^2) \subseteq [0, \|A\|^2]$.
We construct a unitary $U = (1 - i|\lambda|A)\frac{1}{1+i|\lambda|A}$ for $\lambda > 1/\|A\|$. Then we show that $\frac{1-i|\lambda|\alpha}{1+i|\lambda|\alpha} \in \sigma_{\mathcal{A}}(U)$ if and only if $\alpha = \overline{\alpha}$. Then do a long computation.

(e) For all $A \in \mathcal{A}$ and polynomials P, $\sigma_{\mathcal{A}}(P(A)) = P(\sigma_{\mathcal{A}}(A))$.
Prove by factoring.

0.3 Representations

The point of this chapter is to characterise the Hilbert space realisations of a C^* algebra.

1. Homomorphisms preserve positivity

2. The kernel of a homomorphism is a two sided ideal.

3. If $\mathcal{I}$ is a right ideal, an approximate identity $\{E_\alpha\}_\alpha$ is an increasing set of positive operators $0 \leq E_i \leq E_{i+1}$ satisfying the properties

$$\|E_\alpha\| \leq 1, \qquad \alpha \leq \beta \implies E_\alpha \leq E_\beta, \qquad \|E_\alpha I - I\| \to 0.$$

4. Every right ideal has an approximate identity.

   Proof: Partially order every finite family of elements $\{A_1, A_2, \ldots, A_{|alpha|}\}$. Then define $F_\alpha = \sum_{i=1}^{|\alpha|} A_i^* A_i$. The approximate identity will be

$$E_\alpha = |\alpha| F_\alpha \frac{1}{1 + |\alpha| F_\alpha} = 1 - \frac{1}{1 + |\alpha| F_\alpha}.$$

   Eventually the estimate will be, for $A_i \in \alpha$, $(E_\alpha A_i - A_i)^*(A_\alpha A_i - A_i) \leq \frac{1}{|\alpha|}\mathbb{1}$.

5. Every closed two sided ideal is self-adjoint and the factor algebra with the norm $\|\hat{A}\|_{\mathcal{A}/\mathcal{I}} = \inf_{I \in \mathcal{I}} \|A + I\|_{\mathcal{A}}$ is a C^* algebra.

   Proof: $\|A^* E_\alpha - A^*\| = \|A_\alpha - A\| \to 0$, hence $A^* \in \mathcal{I}$ (note that $A^* E_\alpha \in \mathcal{I}$ for all A^*). Then, we adjoint a unit and show C^* properties.

6. Let $\mathcal{A}$ be a unital Banach *-algebra and $\mathcal{B}$ a C^* algebra, and $\pi : \mathcal{A} \to \mathcal{B}$ a *-homomorphism. Then π is continuous with $\pi(A) \leq A$. Additionally, if $\mathcal{A}$ is C^*, then its image is a C^* subalgebra.

   Idea: Compare spectra. If $\lambda\mathbb{1} - A$ is invertible, then $\pi((\lambda\mathbb{1} - A)^{-1})$ is an inverse of $\lambda\mathbb{1} - \pi(A)$. Hence for normal elements we have the desired result. For non normal elements we have $\|\pi(A)\|^2 = \|\pi(A^*A)\| \leq \|A^*A\| = \|A\|^2$.

7. The factor space of the kernel is isomorphic to the image.

   Proof: It is clearly vector space isomorphic. Then, for any invertible π, we have $\|A\| = \|\pi^{-1}(\pi(A))\| \leq \|\pi(A)\| \leq A$ and hence π is norm preserving.

8. A C^* representation is faithful if it satisfies any of the following equivalent conditions

   (a) $\ker \pi = \{0\}$,

   (b) $\|\pi(A)\| = \|A\|$

   (c) $\pi(A) > 0$ for all $A > 0$

   a is obvious, $a \implies b$ by constructing isomorphism onto its image, $b \implies c$ simply because $\|A\| > 0 \implies \|\pi(A)\| > 0$ - note that this is fundamentally a statement about the kernel, and preservation of positivity is obvious and was proven long ago, and $c \implies a$ is clear.

9. Given an invariant closed subspace, we can decompose the representation into a direct sum on it and its orthogonal complement.

10. A representation is called nondegenerate, if the subspace annihilated by it is trivial.

11. A vector is called cyclic if the span of $B\Omega$ is dense in $\mathcal{H}$.

    Cyclic representations are nondegenerate: If we have a nonzero vector $\pi(B)\Omega$ that was annihilated, then we would have $\pi(A)\pi(B)\Omega = \pi(AB)\Omega = 0$ for all A. If we choose $A = \mathbb{1}$ then this means $\pi(B)\Omega = 0$.

    Basically: Because $\pi(\mathbb{1}) = \mathbb{1}$, this is about invariant subspaces.

12. The norm of a C^* algebra is unique.

    If A is normal, $\|A\| = \rho(A)$. Otherwise, $\|A\| = \sqrt{\|A^*A\|} = \sqrt{\rho(A^*A)}$.

13. If $\mathcal{B}$ is a unital C^*-subalgebra of $\mathcal{A}$ (also C^*) and the unit elements coincide, then the spectra of all elements coincide.

14. We define projectors by $P_\Gamma = \frac{1}{2\pi i} \oint_\Gamma \frac{\mathrm{d}z}{z\mathbb{1} - A}$. We can show that this is a projector and projects onto the "eigenspace"

15. The proof involves Hilbert's identity $R(z_1) - R(z_2) = (z_2 - z_1)R(z_1)R(z_2)$ which is proven by $(R(z_2))^{-1} - (R(z_1))^{-1} = (z_2 - z_1)\mathbb{1}$.

16. We define functions of operators by $\hat{f}(A) = \frac{1}{2\pi i} \oint_\Gamma \frac{f(z)}{z\mathbb{1} - A} \,\mathrm{d}z$. This is a homomorphism, i.e. it works on products of functions.

17. If $f : \mathbb{C} \to \mathbb{C}$ is analytic on a neighbourhood of $\sigma_{\mathcal{A}}(A)$, then $f(\sigma(A)) = \sigma(f(A))$.

    If $\lambda \notin \sigma(A)$, then $h_1 = (\lambda - f(z))^{-1}$ is holomorphic on a neighbourhood of $\sigma(A)$. Then $\hat{h}_1(A)$ is an inverse of $\lambda\mathbb{1} - f(\hat{A})$.

    If $\lambda \in f(\sigma(A))$, we define $\lambda = f(z_0)$ and there exists a h_2 holomorphic on a neighbourhood of $\sigma(A)$ with $\lambda - f(z) = f(z_0) - f(z) = (z_0 - z)h_2(z)$. Hence, $\lambda\mathbb{1} - \hat{f}(A) = (z_0\mathbb{1} - A)\hat{h}_2(A)$.

18. $\hat{f} \circ \hat{g} = \widehat{f \circ g}$

0.4 States

1. A linear functional is called positive if $\omega(A^*A) \geq 0$ for all A. If it has norm 1, it is called a state.
2. Examples of states:
   (a) Vector states: $\langle \Omega, \pi(A)\Omega \rangle$
   (b) Statistical matrices: $\sum p_i \omega_i,\ 0 \leq p_i \leq 1, \sum p_i = 1$.
   (c) Density matrices: $\mathrm{Tr}(\rho\pi(A)) = \omega(A)$.
3. Cauchy-Schwarz: $|\omega(A^*B)| \leq \sqrt{\omega(A^*A)}\sqrt{\omega(B^*B)}$.
4. For all ω, the following are equivalent:
   (a) ω is positive
   (b) ω is continuous and $\|\omega\| = \lim_{\alpha\to\infty} \omega(E_\alpha^2)$.

   Additionally, for positive ω, we have
   (a) $\omega(A^*) = \overline{\omega(A)}$
   (b) $|\omega(A)|^2 = \omega(A^*A)\|\omega\|$
   (c) $|\omega(A^*BA)| \leq \omega(A^*A)\|B\|$
   (d) $\|\omega\| = \sup_{\|A\|=1} \omega(A^*A)$
   (e) $\|\omega\| = \lim_{\alpha\to\infty} \omega(E_\alpha)$

   Since this is a very long proof, we wil split it into many parts

   - **(Positivity implies Continuity)** First, we prove that positivity implies continuity and the additional identity.
     We choose a sequence of $\{A_n\}_n,\ 0 \leq A_n,\ \|A_n\| \leq 1$. If for all sequences λ_n we have $\sum \lambda_n A_n \to A \geq 0$, then we also know $\sum_{n=1}^N \lambda_n \omega(A_n) \leq \omega(A) < \infty$. This can only be true for all λ_n if ω are uniformly bounded, i.e. $M_+ = \sup_{\|A\|=1} \omega(A) < \infty$. By decomposition, $\|\omega\| < 4M_+$.
   - **(Alternate Proof)** We alternatively note that $-\mathbb{1}\|A\| \leq A \leq \mathbb{1}\|A\|$. Hence $|f(a)| \leq \|A\| f(\mathbb{1})$. This is generalised in the second equation.
   - **(Proof of the Supplementary Identities)** Since we will need the supplementary identities in the next part, we prove them first. All of them are clear except two, which we shall prove using:
     $$|\omega(A^*BA)|^2 \leq \omega(A*A)\omega(A^*B^*BA), \qquad A^*B^*BA \leq A^*\|B\|^2 A$$
     and
     $$E_\alpha^2 \leq E_\alpha \implies \|\omega\| \leq \lim_{\alpha\to\infty} \omega(E_\alpha^2) \leq \lim_{\alpha\to\infty} \omega(E_\alpha) \leq \|\omega\|.$$
   - **(Proof Sketch: Continuity implies Positivity)** The proof will proceed in three parts. First, we will show that $\omega(\mathbb{1}) = 1$, adjoining an identity if necessary. Then, we will show that $\omega(A)$ must be real for A self adjoint. Finally, we will show the desired result.
   - **(Proof that $\omega(\mathbb{1}) = 1$)** We show that $\lim_{\alpha\to\infty} E_\alpha^2 = 1$. Then, we have $\omega(\mathbb{1}) = \lim_{\alpha\to\infty} \omega(E_\alpha^2) = 1$.
     In the non unital case, we adjoin a unit, then prove that the extension $\tilde{\omega}((\lambda, A)) = \lambda + \omega(A)$ that $\|\tilde{\omega}\| = 1$. We do this by $|\tilde{\omega}(\lambda\mathbb{1} + A)| = |\lambda + \omega(A)| = \lim_{\alpha\to\infty} |\lambda\omega(E_\alpha^2) + \omega(AE_\alpha^2)| = \limsup_{\alpha\to\infty} \|\lambda E_\alpha^2 + AE_\alpha^2\|_{\mathcal{A}} \leq \sup_{\|B\|=1} \|\lambda B + AB\| = \|\lambda\mathbb{1} + A\|_{\overline{\mathcal{A}}}$, and thus $\|\tilde{\omega}\| \leq 1$. However, since $\tilde{\omega}(\mathbb{1}) = 1$, and thus $\|\tilde{\omega}\| = 1$.

- **($\omega(A)$ is real if A is self adjoint)** First, we assume that $\omega(A) = \alpha + i\beta$. Then, we consider $A + i\gamma\mathbb{1}$. Since the spectrum of A is real and $A + i\gamma\mathbb{1}$ is normal, then $\|A + i\gamma\mathbb{1}\| = \rho(A + i\gamma\mathbb{1}) = \sqrt{\|A\|^2 + \gamma^2}$. Now, $|\omega(\alpha + i\gamma\mathbb{1})| = |\alpha + i(\beta + \gamma)| = |\alpha + i(\beta + \gamma)| = \sqrt{\alpha^2 + (\beta + \gamma)^2} \geq |\beta + \gamma|$.

  Hence, $|\beta+\gamma| \leq \sqrt{\|A\|^2 + \gamma^2}$ for all γ. If we choose $\gamma = \delta\|A\|^2\beta/|\beta|$, we find that $\beta^2+2\delta|\beta|\|A\|^2 \leq \|A\|^2$. Sending $\delta \to \infty$, we find that $\beta = 0$.

- **(Final Steps)** Since $A^*A \geq 0$,

$$\|\mathbb{1} - \frac{A^*A}{\|A^*A\|}\| = \|1 - \frac{A^*A}{\|A\|^2}\| \leq 1.$$

  We then have

$$\left|\omega(\mathbb{1}) - \frac{\omega(A^*A)}{\|A\|^2}\right| \leq 1$$

  and

$$\left|1 - \frac{\omega(A^*A)}{\|A\|^2}\right| \leq 1.$$

  This shows that ω is positive.

5. The sum of two positive linear functionals $\omega_1,\ \omega_2$ is positive and $\|\omega_1 + \omega_2\| = \|\omega_1\| + \|\omega_2\|$.

   Proof: $\lim(\omega_1(E_\alpha^2) + \omega_2(E_\alpha^2)) = \lim \omega_1(E_\alpha^2) + \lim \omega_2(E_\alpha^2) = \|\omega_1\| + \|\omega_2\|$. Hence, the states form a convex subset.

6. We can define a partial order: $\omega_1 \leq \omega_2$ if $\omega_2 - \omega_1$ is positive.

7. We call a state ω pure, if $\eta \leq \omega$ implies that $\eta = \lambda\omega$.

8. **(GNS Construction)**

CHAPTER ONE

C^* Algebras

Theorem 1.1. If we have an algebra $\mathcal{A}$, there is an algebra $\bar{\mathcal{A}}$ containing $\mathcal{A}$ as a sub algebra that has an identity. If $\mathcal{A}$ is C^*, then so is $\overline{\mathcal{A}}$.

Definition 1.2. The set of λ such that $\lambda 1 - A$ is invertible is called the resolvent set of A. Its complement in the complex plane is called the spectrum.

Definition 1.3 (Von-Neumann Series)**.** We can expand the resolvent using the geometric series

$$\frac{1}{\lambda - A} = \frac{1}{\lambda}\frac{1}{1 - A/\lambda} = \frac{1}{\lambda}\sum_{n=0}^{\infty}\left(\frac{A}{\lambda}\right)^n.$$

Remark 1.4. This shows that for $\lambda > \|A\|$, the series is absolutely convergent. This implies that the spectrum is bounded.

Theorem 1.5. The spectral radius is given, for general A, by

$$\rho(A) = \inf_n \|A^n\|^{1/n} = \lim_{n\to\infty} \|A^n\|^{1/n} \leq \|A\|$$

Proof. The former follows from the root test. □

Theorem 1.6. Let $\mathcal{A}$ be a unital C^* algebra.

1. If A is normal then
$$\rho(A) = \|A\|$$
2. If A is an isometry, then
$$\rho(A) = 1$$

3. If A is unitary, then
$$\sigma_A \subseteq S^1$$
4. If A is self adjoint, its spectrum is real
5. For all polynomials P,
$$\sigma(P(A)) = P(\sigma(A))$$

Proof. 1. We use the C^* property to move the power out of the norm:

$$\begin{aligned}\|A^{2^n}\|^2 &= \|(A^*A)^{2^n}\| \\ &= \|(A^*A)^{2^{n-1}}\|^2 \\ &= \|A^*A\|^{2^n} \\ &= \|A\|^{2^{n+1}}\end{aligned}$$

2. We have

$$\begin{aligned}\|A^{2n}\|^2 &= \|(A^*)^{2n}A^{2n}\| \\ &= \|1\| = 1\end{aligned}$$

3. We have
$$\sigma(A) = \overline{\sigma(A^*)} = \overline{\sigma(A^{-1})} = \left(\overline{\sigma(A)}\right)^{-1}.$$
□

Theorem 1.7 (Uniqueness)**.** The norm of a C^* algebra is unique.

Proof. The norm is given by, for normal elements,

$$\|A\| = \rho(A).$$

For not normal elements, we have
$$\|A\| = \sqrt{\|A^*A\|} = \rho(A^*A).$$
□

CHAPTER TWO

Homomorphisms

Theorem 2.1. Homomorphisms preserve positivity

Theorem 2.2. The kernel of a homomorphism is a 2 sided ideal.

Definition 2.3 (Approximative Identity)**.** If $\mathcal{I}$ is a right sided ideal, an approximate identity is defined such that

$$\|E_\alpha I - I\| \xrightarrow{\alpha\to\infty} 0.$$

Theorem 2.4. Every right ideal possesses an approximate identity.

Proof. The idea is: We partially order the set of all finite families $\alpha = \{A_1, \ldots, A_{|\alpha|}\}$, then we construct

$$F_\alpha = \sum_{i=1}^{|\alpha|} A_\alpha A_\alpha^*.$$

Then, by adjoining a unit to $\mathcal{A}$ if necessary, we can define

$$E_\alpha = |\alpha| F_\alpha \frac{1}{1+|\alpha|F_\alpha} = 1 - \frac{1}{1+|\alpha|F_\alpha}.$$

The proof idea is showing that $\alpha \to \infty$, hence this looks like a 1. □

Theorem 2.5. Every closed two sided ideal is self adjoint and the factor algebra is a C^* algebra.

CHAPTER THREE

Quantum Mechanics

Definition 3.1 (Symplectic Vector Space)**.** A symplectic vector space is a real vector space with a nondegenerate antisymmetric bilinear map. Nondegenerate means that if

$$\theta(u, v) = 0$$

for all $u \in V$, then $v = 0$.

Definition 3.2 (Weyl System)**.** A Weyl system of a symplectic vector space is a map $V \to \mathcal{A}$ such that

$$\begin{aligned} W(0) &= 1 \\ (W(\phi))^* &= W(-\phi) \\ W(\phi)W(\psi) &= e^{-\frac{i}{2}\theta(\phi,\psi)}W(\phi+\psi) \end{aligned}$$

Definition 3.3 (Weyl Algebra)**.** We define a particular vector space by letting $V = \mathbb{R}^2$, and defining the symplectic product as

$$\theta((\xi_1, \eta_1), (\xi_2, \eta_2)) = \eta_1\xi_2 - \xi_1\eta_2.$$

The algebra generated by $W(\xi, 0),\ W(0, \eta)$ is called the Weyl algebra.

Theorem 3.4. Let $(\mathcal{A}, W)$ be a Weyl system of a symplectic vector space (V, θ). Then

1. $W(\phi)$ is unitary for all $\phi \in V$
2. $\|W(\phi) - W(\psi)\| = 2$ for all $\phi \neq \psi \in V$.
3. $\mathcal{A}$ is not separable, unless $V = \{0\}$.
4. The family $\{W(\phi)\}_{\phi \in V}$ is linearly independent.

Proof. 1. We have

$$\begin{aligned}[W(\phi)]^* W(\phi) &= W(-\phi)W(\phi) \\ &= e^{-\frac{i}{2}\theta(-\phi,\phi)} W(0)\end{aligned}$$

Because θ is antisymmetric, it follows that $\theta(-\phi,\phi) = -\theta(\phi,\phi) = 0$.

2.

□

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